

## Description of files in this folder

1. run\_trim\_galore.sh
  - a. Command used to run TrimGalore on the samples
  - b. FASTQs are available under NCBI BioProject ID PRJNA1328687
2. run\_kallisto.sh
  - a. Command used to run kallisto after trimming
3. sample\_info\_mouse\_brain\_rnaseq.xlsx
  - a. Metadata used for the analyses below
4. pca\_and\_deseq2.Rmd
  - a. Inputs:
    - i. The Kallisto output files which are available on GEO (GSE341280)
    - ii. The t2g.txt that came with the standard mouse transcriptome index that was used by kallisto for pseudoalignment
      1. Downloaded from the following link on March 5, 2025:
        - a. [https://github.com/pachterlab/kallisto-transcriptome-indices/releases/download/v1/mouse\\_index\\_standard.tar.xz](https://github.com/pachterlab/kallisto-transcriptome-indices/releases/download/v1/mouse_index_standard.tar.xz)
    - iii. sample\_info\_mouse\_brain\_rnaseq.xlsx (metadata, available in this folder, see above)
  - b. Outputs:
    - i. PCAs for each mouse brain RNA-seq experiment (there were two, see Methods) – used for a figure
      1. expt1\_samples\_pca.pdf
      2. expt2\_samples\_pca.pdf
    - ii. hemoglobin\_genes\_plot.pdf – used for a figure
    - iii. cooks\_distance\_tmao\_samples.pdf – used for a figure
    - iv. DESeq2 results, used to create a supplementary table
      1. Contained within deseq2\_outputs subfolder
    - v. Volcano plots for each metabolite tested (TMAO, PA, IAA, GDCA) – used for a figure
5. fgsea.Rmd
  - a. Inputs:
    - i. The DESeq2 results (produced by the previous script) for TMAO and GDCA
    - ii. The following collections of gene sets, downloaded on July 4, 2025 from: <https://www.gsea-msigdb.org/gsea/msigdb/mouse/collections.jsp>
      1. Hallmark gene sets – “mh.all.v2025.1.Mm.symbols.gmt”

2. REACTOME subset of CP (canonical pathways) –  
“m2.cp.reactome.v2025.1.Mm.symbols.gmt”

- b. Outputs:
  - i. All contained within the fgsea\_results subfolder:
    1. Results of gene set enrichment analysis, using the input collections of gene sets
      - a. Used for a supplementary table
    2. PDFs depicting selected significant terms from GSEA
      - a. Used for figures
6. marker\_gene\_analysis\_tmao.Rmd
  - a. Inputs:
    - i. Processed single-cell data from <https://doi.org/10.1038/s41593-019-0491-3>, downloaded from [https://singlecell.broadinstitute.org/single\\_cell/study/SCP263/aging-mouse-brain](https://singlecell.broadinstitute.org/single_cell/study/SCP263/aging-mouse-brain)
    - ii. Genes significantly regulated by TMAO, as outputted by the DESeq2 script above
  - b. Outputs:
    - i. Plot showing % of marker genes downregulated by TMAO, for each of six brain cell classes
      1. Used for a figure
    - ii. RDS containing the computed marker genes per class
    - iii. TSV containing the computed marker genes per class, used for a supplementary table
    - iv. A TSV for a source data file
7. marker\_gene\_analysis\_gdca.Rmd
  - a. Inputs:
    - i. The RDS outputted by the previous script
    - ii. Genes significantly regulated by GDCA, as outputted by the DESeq2 script above
  - b. Outputs:
    - i. Plot showing % of marker genes downregulated by GDCA, for each of six brain cell classes
      1. Used for a figure
    - ii. A TSV for a source data file
8. gaba\_a\_receptor\_genes\_heatmap.Rmd
  - a. Inputs:
    - i. Same as those for pca\_and\_deseq2.Rmd, above
  - b. Outputs:

- i. A heatmap, used for a figure, showing downregulation of GABA<sub>A</sub> receptor subunits in the brains of TMAO-treated mice
- 9. star\_alignment\_script.sh
  - a. Bash script that was used to run STAR alignment (instead of kallisto) on the trimmed fastq files, in order to prepare for differential splicing analysis with rMATS-turbo
- 10. star\_index\_script.sh
  - a. Bash script that was used to create the STAR index for STAR alignment
- 11. rmats\_turbo\_script.sh
  - a. Bash script that was used to run rMATS-turbo after STAR alignment
- 12. differential\_splicing\_events.Rmd
  - a. Inputs:
    - i. Files outputted by rMATS-turbo, available in this folder within the subfolder rmats\_output
  - b. Outputs:
    - i. Excel files
      - 1. diff\_splicing\_analysis\_results.xlsx
      - 2. sig\_diff\_splicing\_events.xlsx (used for a supplementary table)
    - ii. Volcano plot used for a figure panel
      - 1. Along with a source data file
- 13. sashimi\_code subfolder
  - a. Contains files used to generate supplementary sashimi plots of differential splicing in *Rsrp1*, *Cpsf4*, *Ythdf3*, and *Kmt2b* in brains from TMAO- vs. saline-treated mice.
  - b. Specifically contains:
    - i. Files containing the commands used to run ggsashimi.py (which was downloaded from <https://github.com/guigolab/ggsashimi> on June 26, 2025)
    - ii. Input files required for running those commands:
      - 1. palette\_saline\_grey.txt
      - 2. input\_bams.tsv
    - iii. Raw output PDFs of the ggsashimi commands, which were later edited in Illustrator to reduce visual clutter and emphasize the junctions relevant to each differential splicing event
  - c. Required files not included in this repo:
    - i. Bam files (generated by STAR – very large)
    - ii. gencode.vM37.primary\_assembly.sorted.gtf
      - 1. This file was generated from gencode.vM37.primary\_assembly.annotation.gtf, which was

downloaded from GENCODE (release M37; GRCm39;  
<https://www.gencodegenes.org/mouse/>).

2. The annotation file was sorted on the command line using  
“sort -k1,1 -k4,4n”